

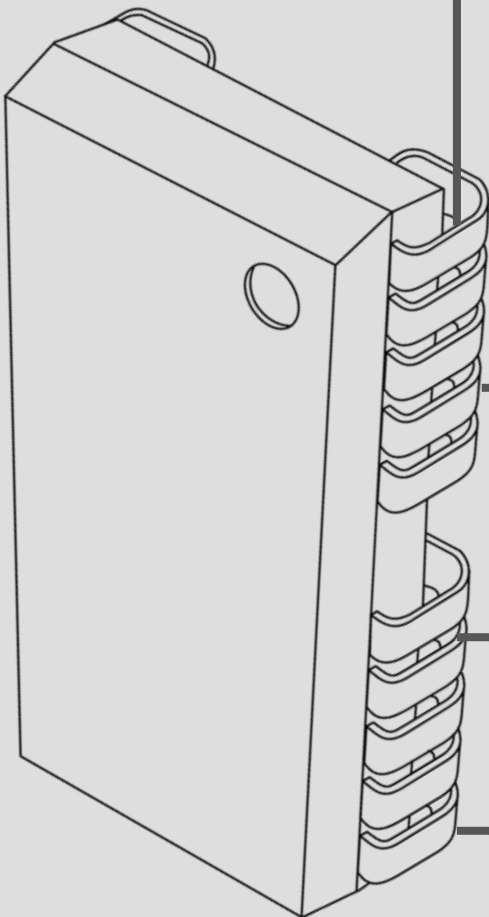
Sensor Integration And Testing

Integrating sensors into a formerly manual robot became one of the most challenging and rewarding parts of VoltEdge's autonomous journey. Since the BaseBot lacked sensing hardware, every capability had to be built from scratch. After acquiring the IR line array, gyroscope, and ultrasonic sensor, our team spent days translating these components into a comprehensible system. Loose wires and raw readings gradually transformed into a coordinated sensing network guiding the robot autonomously.

The first major step was IR line tracking. We placed a black tape strip on the floor as a test track. Dozens of attempts were made—sometimes the robot followed smoothly, other times it drifted or overcorrected. Each attempt taught us something new. We adjusted thresholds, correction logic, and sensor spacing until the robot confidently read the tape, evolving the IR array into a reliable guide.

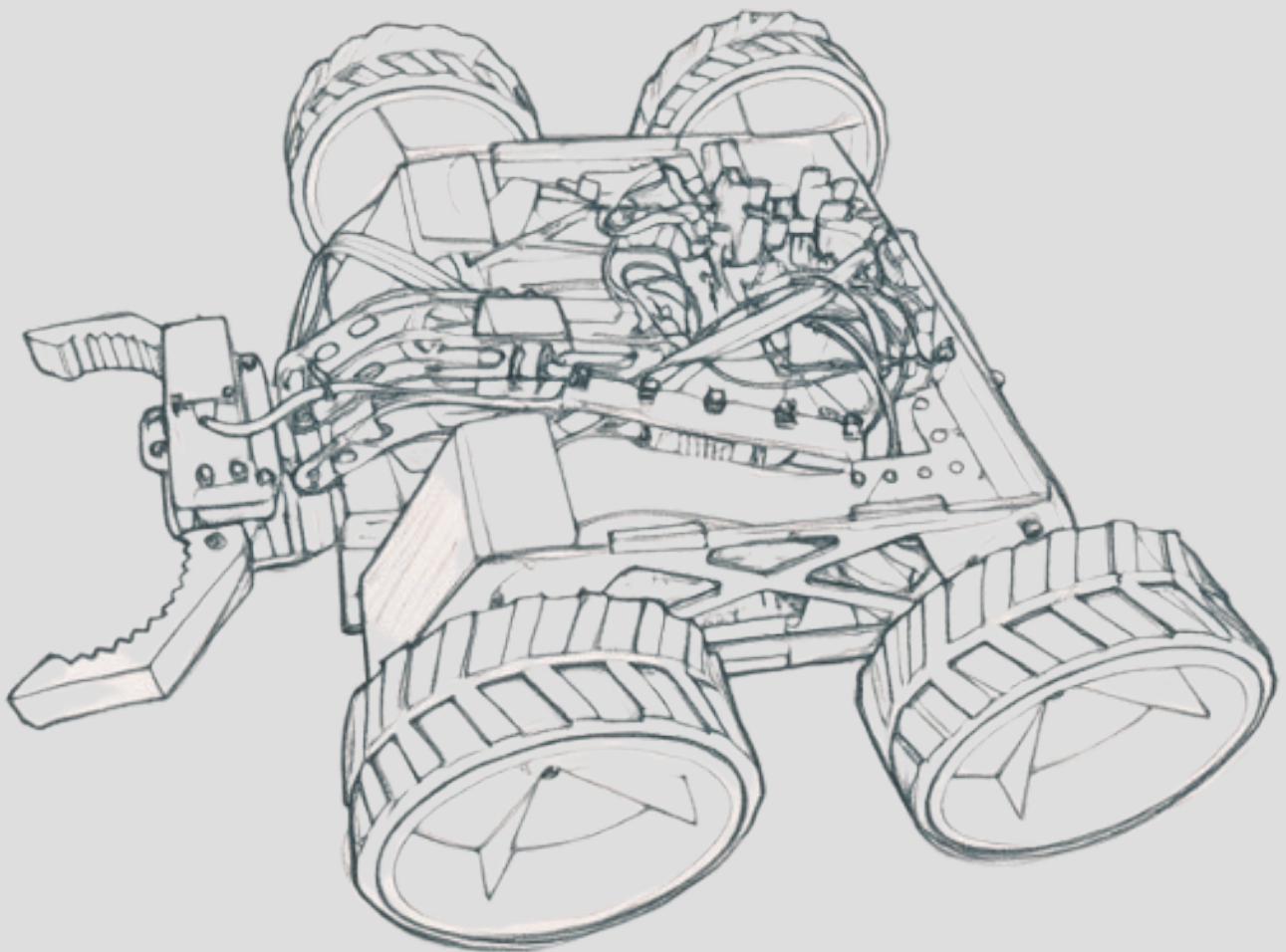
The gyroscope was the second key element. Initially, the robot rotated slightly or took inconsistent turns. Integrating the gyro stabilized orientation, corrected drift, and ensured precise angle turns. This improved straight-line accuracy and predictability.

For distance detection, the ultrasonic sensor was tuned to recognize the charge from the right distance. We tested multiple ranges until the robot consistently aligned correctly, giving it the “awareness” needed to approach objects without collisions.



Throughout this process, VoltEdge received valuable guidance from Danish Gawai Sir, who helped us understand sensor behaviour and refine our autonomous logic over several days. His mentorship, combined with our repeated testing, helped transform raw data into usable decisions the robot could rely on.

By the end of integration, the robot responded not just to code—but to the environment. It followed the black tape confidently, held direction steadily, and slowed down precisely near the charge. These tests were where the robot learned, where we learned, and where autonomous mode truly came to life.



Strategy

Basebot Operational Strategy As Per Guide

The BaseBot was designed with a simple, predictable strategy in mind — a strategy that every team could understand before attempting modifications or advanced maneuvers. According to the NRL guide, the BaseBot's primary purpose is to drive reliably, collect charges, and score them with consistency, using a stable differential drivetrain and a straightforward servo-based arm. Its four-wheel tank-drive system gives it strong traction and smooth directional control, allowing it to move quickly across the field, push charges when needed, and adjust its position during scoring attempts. The gripper mechanism, mounted on the front servo arm, is designed to lift a single charge securely and place it into the scoring zone.

The guide emphasizes simplicity: the BaseBot should first master basic collection cycles. This means approaching a charge cleanly, lifting it with the arm, backing out safely, and driving toward the drop zone without losing grip. Each cycle is meant to be smooth, stable, and repeatable. The robot's low height also allows it to pass under the Charge Station for full parking points, giving teams an additional strategic advantage during endgame situations.

The strategy proposed in the guide encourages teams to focus on control, predictability, and consistency before attempting any complex maneuvers or modifications. Its drivetrain is meant to provide straight, balanced motion, while the gripper's geometry is designed to hold the charge with minimum errors. In simple terms, the BaseBot's operational strategy is built around mastering the fundamentals: drive straight, collect confidently, score cleanly, and repeat.

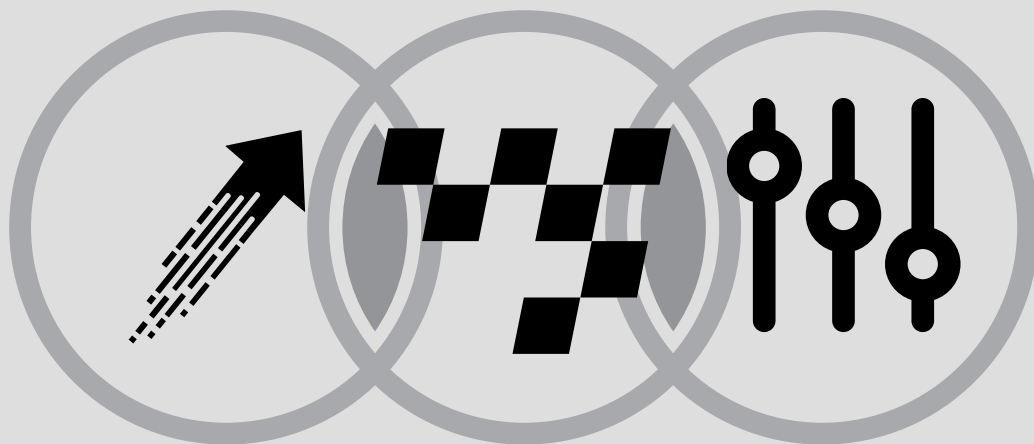
For VoltEdge, this base strategy became the foundation we learned from before evolving into customized controls, autonomous improvements, and advanced match-planning. The BaseBot taught us the rhythm of the arena — one reliable cycle at a time.

Our Team's Match Play Strategy, and Adjustments

For VoltEdge, match play wasn't about taking risks — it was about taking the smartest, shortest, and most reliable path to score. From the very beginning of our testing sessions, we realised that the BaseBot responded best when its movement stayed simple, direct, and controlled. This became the foundation of our match strategy: reduce unnecessary turns, avoid long routes, and always choose the shortest cycle path to the charge and back to the drop zone.

With Omer as the driver, the robot's customised control mapping allowed quick reactions and smooth handling. The left joystick kept the robot moving cleanly in straight lines, the right joystick handled tight directional adjustments, and the redesigned trigger layout gave instant control over the arm and gripper. This meant that our cycles could be fast without sacrificing accuracy.

Our Strategy Inside The Arena Is Built On Three Principles:

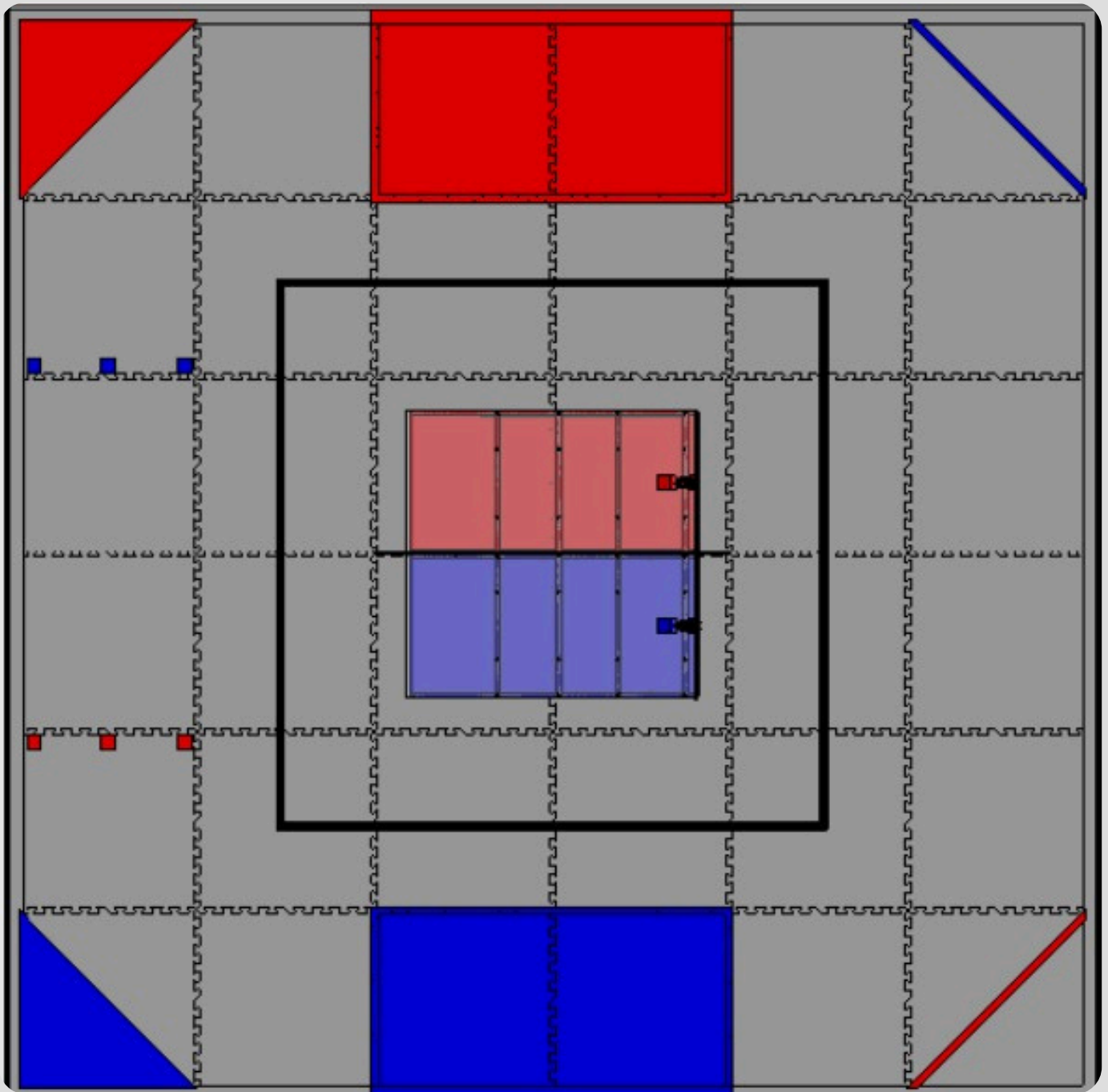


1. Shortest Path First —
Instead of navigating wide arcs or complicated turns, we always pick the most direct route from spawn point → charge → drop zone. The robot accelerates, collects, scores, and resets as quickly as possible.

2. Stable Driving Over Speed—
Even though the BaseBot can move fast, we prioritise clean control. A stable entry into the charge zone, a steady lift, and a controlled return to scoring always outperform unpredictable speed.

3. Instant Adjustments —
Thanks to our custom controls, small joystick inputs allow rapid micro-adjustments. If the robot drifts slightly or a charge is placed at an angle, the driver can correct the alignment immediately.

During testing, we refined this strategy repeatedly — shortening paths, reducing unnecessary movement, and timing each cycle until it became second nature. Our focus remained on consistency, not complexity. A reliable routine, performed over and over with precision, became our competitive advantage.



Every movement inside the arena is shaped by one VoltEdge principle: If there is a shorter route, we take it.

This approach keeps our cycles fast, our scoring consistent, and our strategy simple enough to execute under pressure — giving EdgeBot the best chance to perform confidently when it matters most.

Outreach

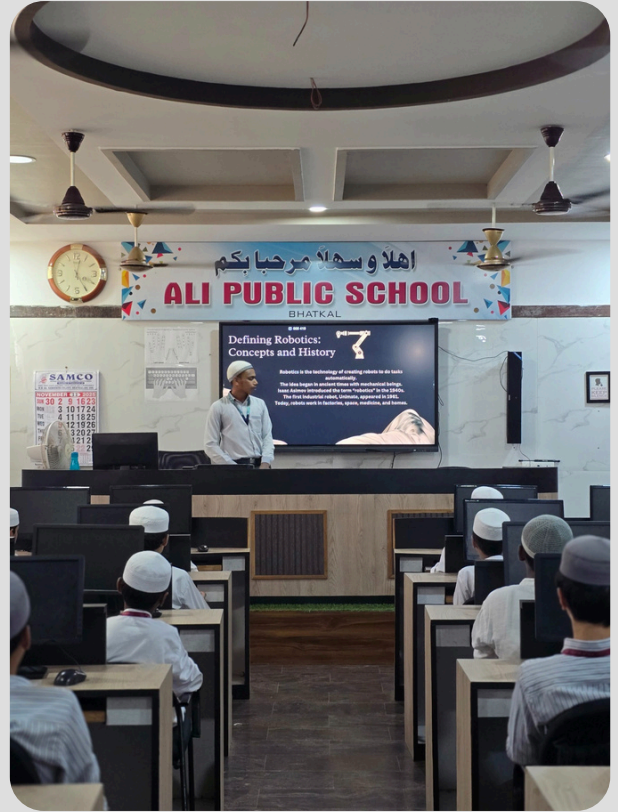
Workshops, And Events Conducted.

One of the most meaningful parts of VoltEdge's outreach journey was the workshop we conducted at Ali Public School on 6th November. As students of Grade 12 balancing midterms, robotics work, and council responsibilities, organising this session was challenging — but it became one of the most impactful moments of our NRL season. The workshop introduced nearly forty students to the world of robotics, turning complex concepts into something they could understand and get excited about.

The session opened with Mohiddin, whose hosting guided the entire flow — welcoming teachers, connecting each segment, and keeping the atmosphere comfortable for students who were experiencing robotics for the first time. Zaid took the stage next, explaining the evolution of robotics from Isaac Asimov's ideas to the robots used today, giving students a timeline of how technology has grown. Omer followed with a simple yet engaging explanation of sensors, helping students understand how robots sense and interact with their surroundings.

To make the learning interactive, Mohiddin led a quiz that tested curiosity and rewarded active participation. After this, Abdullah Saami introduced students to practical robotics concepts, the upcoming hands-on Robotics course, and even showed them the official NRL website and VoltEdge's robot video — making the idea of competing at IIT Bombay feel real and achievable. Finally, Irfan closed the session with a short motivational talk that encouraged students to stay curious and keep exploring engineering.

The workshop concluded with photos, small gifts, and documentation by Shamveel, who captured every key moment through video and pictures. This event became an important part of VoltEdge's outreach mission — not just teaching robotics, but showing students that innovation can begin anywhere, even in a small classroom, even in one afternoon.





On 15th November, Team VoltEdge launched its hands-on Basic Robotics Course at Ali Public School, marking the beginning of a structured learning journey for young students. The course was designed to introduce them to the fundamentals of robotics — understanding components, sensing mechanisms, wiring basics, and simple movement logic — through practical, student-friendly sessions. To ensure every participant could learn by doing, we distributed a Basic Robotics Kit that included essential components for building and experimenting. The goal of the course is not just to teach robotics, but to spark curiosity, build technical confidence, and show students that engineering becomes meaningful the moment they touch, build, and create with their own hands.



Community Involvement, And STEM Promotion.

For VoltEdge, robotics was never meant to stay within our own workspace — it was meant to reach the community around us. From the moment we registered for NRL, we knew that our journey would mean much more if it inspired students who had never seen robotics up close. This vision led to our STEM outreach efforts, where we shared what we learned with younger students and opened a door for them to explore science and engineering.

Our biggest step in community involvement came through the workshop at Ali Public School, where nearly forty students experienced robotics for the first time. We didn't just demonstrate a robot — we explained how technology evolves, how sensors work, how robots think, and how students themselves could start learning with simple circuits. The session became a hands-on introduction to STEM, turning robotics from something “difficult” into something exciting and achievable.

Beyond teaching, the workshop helped build curiosity. Students engaged in quizzes, asked questions, interacted with components, and saw real robot footage from VoltEdge — connecting theoretical concepts to what a real team from their own town was building. This exposure created awareness about engineering pathways, national competitions like NRL, and how STEM fields can shape their future.

Our outreach did not end in one session. Through the upcoming Hands-On Robotics Course at the same school, VoltEdge is continuing to guide students step-by-step — from basic electronics to building small working robots for their science exhibition. This initiative gives younger learners a long-term foundation in STEM and encourages them to pursue engineering with confidence.

Every effort we put into community involvement aims toward one goal: making robotics accessible, inspiring, and real for the next generation.

VoltEdge believes that sharing knowledge is just as important as building a robot — because every mind we spark today might become tomorrow's innovator.

Team VoltEdge: Field Engagement & Tech Outreach Report

Remote Drone-Assisted Research & Innovation Concepts

As part of our commitment to real-world engineering, Team VoltEdge expanded its outreach through a deep dive into agricultural automation. Our research team began exploring the idea of a remote drone-assisted harvesting system, a concept born from long hours of field observation and conversations with farmers who struggle with the danger and difficulty of harvesting fruits from tall trees. The team is now actively working on developing this harvesting drone — a flying tool designed to identify ripe fruits, approach safely, and support farmers where ladders and manual effort fall short. It represents the VoltEdge approach perfectly: innovation that rises from real problems and aims to make real impact.

In the same competition, another innovation emerged from within the team. Irfan, driven by curiosity and a focus on weather challenges faced by farmers, proposed his own research concept — an automated field shutter system. His design used rain and wind sensors to gather real-time environmental data and instantly deploy a protective shutter whenever storm conditions threatened crops. Instead of relying on manual intervention or guessing weather patterns, his system acted automatically, shielding farmland from sudden rain bursts, strong winds, and storm damage.

Together, these concepts showcased the spirit of VoltEdge — different minds solving different problems, yet all driven by the same purpose: using engineering to improve lives. Whether it's a drone that climbs where farmers cannot, or a shutter that responds faster than human reaction, every idea reflects our belief that innovation must be practical, thoughtful, and designed to serve the world beyond the workshop.

In the same competition, Omer and Zaid's drone-harvesting concept earned them the Aspire Scientist Award, recognising their innovation and clarity of research.

Impact For Robotics And NRL

The influence of robotics—and specifically the National Robotics League—extended far beyond our own team. For many students in our community, NRL became their first introduction to a world where engineering feels alive, creative, and achievable. Through our workshop and ongoing robotics course, students experienced robotics not as a distant concept found in textbooks, but as something they could build with their own hands.

For younger students who attended our session at Ali Public School, seeing a real NRL team from Bhatkal changed their perception immediately. Robotics stopped feeling “too advanced” or “only for big cities.” Instead, they saw that students from their own town, from their own background, could compete on a national stage in IIT Bombay. This made STEM feel accessible—something they could realistically grow into.

Our demonstrations of sensors, circuits, and real robot footage showed students how engineering could solve problems in daily life. Many of them expressed interest in joining robotics in the future; some even approached us afterward, asking how they could participate next year. By showing them the official NRL website, explaining the challenge, and presenting VoltEdge’s own robot video, we connected them directly to a national STEM opportunity.

The upcoming Hands-On Robotics Course strengthened this impact even more. Students now see a path—not just inspiration. They know they can move from learning basic circuits to building small robots for their science exhibition, and eventually step into competitions like NRL.

For our town, NRL became more than a competition.

It became:

- a spark for student curiosity
- a platform that made STEM careers feel real
- a source of confidence for young learners
- a reminder that innovation can begin anywhere

VoltEdge didn’t only build a robot.

We helped build interest, awareness, and belief.

The impact of robotics and NRL will continue to grow with every student who decides to explore engineering because of what they saw, learned, and experienced through our outreach

Social Media And Branding

Branding Based On Team Identity

VoltEdge's branding didn't start with colours or logos — it started with who we were as a team. From the first day we registered, we knew our identity had to reflect the energy, precision, and vision that shaped every step of our journey. That identity became the core of our branding: gold, yellow, black, and our signature message — Driven by Volts, Defined by Vision. These colours weren't random choices. Gold symbolised ambition and excellence, yellow represented innovation and learning, while black gave us the clean, sharp foundation that made everything look modern and focused.

Our team name, VoltEdge, captured our personality perfectly. “Volt” represented power, energy, and electronics — the heart of robotics. “Edge” reflected precision, competitiveness, and the sharpness of engineering design. Together, the name told the story of a team that builds with intensity and thinks with clarity. This identity was strengthened through our official team logo, which stayed consistent across posts, videos, documents, and the engineering portfolio. We never altered its colours, because the logo became a symbol of our unity and professionalism.

Every part of our branding was chosen with intention. Our Instagram posts, reveal videos, behind-the-scenes clips, and workshop documentation followed the same colour style — dark backgrounds with sharp golden and yellow accents. Even the Engineering Portfolio cover page and Table of Contents were built around this signature look. Whether someone saw our robot, our graphics, or our uniform visuals, they immediately recognised VoltEdge.

The goal of our branding was simple:
to build a team identity strong enough that even before the robot enters the arena, people know who built it.

For VoltEdge, branding is not decoration — it is identity, discipline, and representation. To build a unified presence on and off the field, we designed and customised our own team apparel, which includes our official VoltEdge team jersey and VoltEdge team jacket. Both pieces were crafted in our signature black-and-gold palette, carrying our logo and tagline with precision. This consistent visual identity strengthens how we present ourselves in workshops, competitions, social media content, and official events, ensuring that wherever VoltEdge stands, our presence is instantly recognisable.



VoltEdge's brand reflects who we are — focused, disciplined, modern, and driven by a vision larger than the robot itself.

Social Media Strategy.

For Team VoltEdge, social media was never just a place to post — it became our parallel arena.

While the robot was being assembled piece by piece, our identity was being built online, post by post.

Between 29 October and 28 November, VoltEdge created a consistent digital presence on Instagram that focused on three pillars:

1. Visibility



2. Engagement



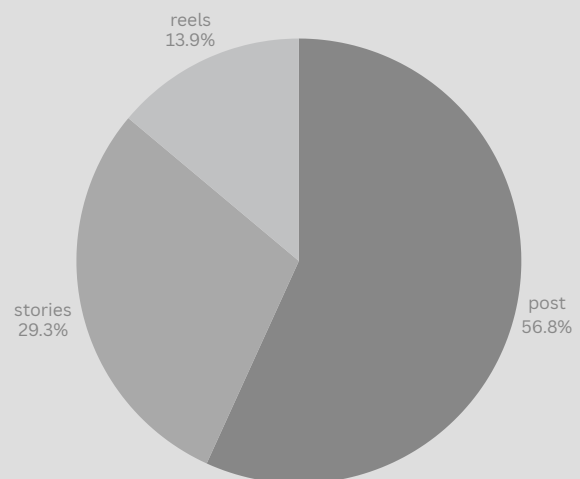
3. Community Awareness



Even as a newly-formed team, our page began gaining traction far beyond what we expected.

In just one month, the page recorded:

- 68,385 total views (As of 29Nov)
- 13,603 accounts reached
- 85.3% of viewers were non-followers
- 56.8% of views came from reels
- 29.3% from posts
- 13.9% from stories
- 1,883 profile visits (a +969.9% growth from the previous month)



This meant that almost all of our visibility came from outside our circle — proving that our content wasn't just “seen”... it was noticed.

Even October (before VoltEdge was fully active) gave us an early indicator of potential:

- 12.9K total views
- 80% views from non-followers
- 37 followers gained that month

By November, that momentum transformed into our strongest month so far.

Content Flow And Posting Style

Our content strategy was intentionally simple, clean, and aligned with NRL's weekly challenge rhythm.

We didn't chase trends — we posted what represented VoltEdge:

- Team Packing & Chaos

A raw, relatable look into the behind-the-scenes reality — wires on the floor, last-minute fixes, and all the laughter in between.

- Apparel Reveal

Aesthetic clips showcasing VoltEdge merch and identity — black, gold, bold.

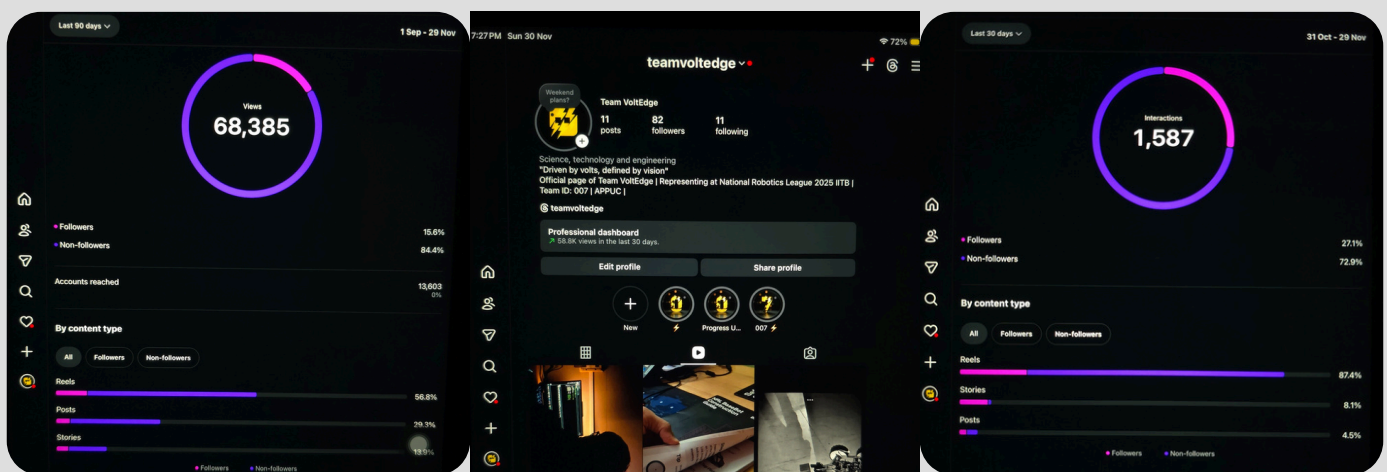
- Workshop Highlights

Special moments from the Ali Public School workshop, capturing students' reactions and curiosity.

- Memes & Relatable BTS Moments

Our top reel — 20,000+ views — was a simple meme of the team watching other teams make reels.

It showed that authenticity always wins.



Strategic Direction Going Forward

Based on our current data and NRL's weekly challenge flow, VoltEdge continues focusing on:

- High-reach reels (build clips, meme edits, robot POV shots)
- Team-identity posts (logo, apparel, reveal teasers)
- Workshop and community updates
- Consistent brand colors (black + gold)
- Clear captions & NRL hashtags to stay aligned with competition visibility
- Posting remaining high-value content:
 - Team Apparel Reveal
 - Team Packing Chaos
 - Workshop Highlights
 - Robot Reveal Video (next to be posted)

VoltEdge Social Media Strategy Summary

In simple words:

We didn't just post.

We built awareness.

We built identity.

And without a single paid promotion, VoltEdge reached nearly 65,000 people in one month.

This is the story our analytics tell —

a team that started small, grew consistently, and positioned itself strongly even before entering the real arena.



@teamvoltedge

Mandatory Upload :Team And Robot Reveal Video

For VoltEdge, the reveal videos weren't just compulsory NRL uploads — they were the public face of our identity. These two videos marked our first official appearance as Team VoltEdge and the first introduction of our robot, EdgeBot, to the world.

Team Reveal Video — Crafted in the Middle of Chaos

The team reveal challenge was announced on 6 November, right when half the team was preparing for an educational tour scheduled that very night. With the clock ticking and only a few hours left before departure, we made the decision to shoot the team reveal early in the morning of 7 November on our school campus.

While the rest of the team left for their tour at 11:15 PM, the responsibility of editing and completing the video fell entirely to Abdullah Saami Sada. Throughout the tour week (7–13 November), Abdullah handled the full post-production alone — stitching clips, refining transitions, matching VoltEdge's brand colors, and ensuring the final video met NRL standards.

Despite the tight schedule, the team reveal went live on time, marking the official digital birth of VoltEdge.

Robot Reveal Video — Scheduled and Strategically Released

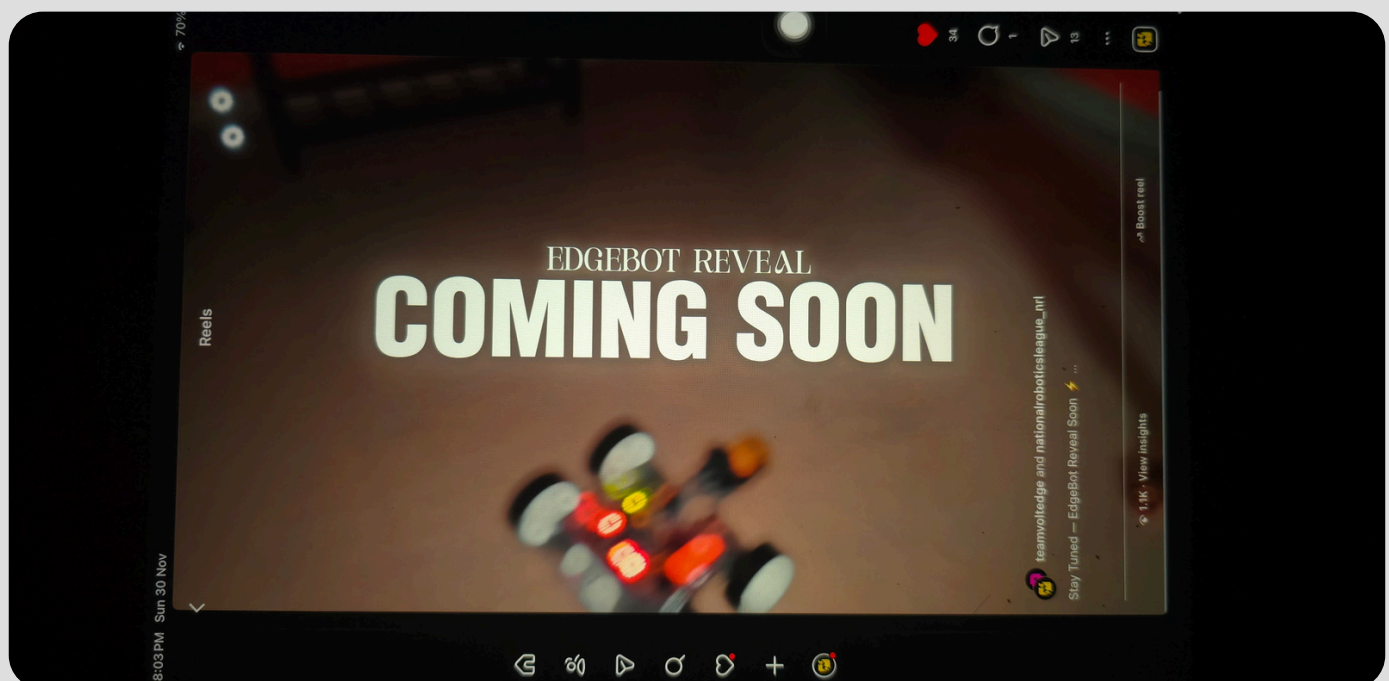
The robot reveal carries more weight — it showcases EdgeBot for the first time and sets the tone for our engineering identity.

Our posting plan is structured and intentional:

- 30 November — Uploaded the Robot Reveal Trailer, giving the audience a cinematic peek at EdgeBot.
- 4 or 5 December — Post the Full Robot Reveal Video, right before portfolio submission and just days before the NRL competition.

The reveal is timed to maximize visibility, maintain engagement, and keep VoltEdge active during the critical pre-event period when teams across India are preparing to travel to IIT Bombay.

Both reveal videos are more than submissions — they are a reflection of our journey, our coordination, and our commitment to presenting VoltEdge with clarity, style, and discipline



Vision For Change

System Design, And Problem Analysis.

The inspiration for our innovation came from observing a simple but dangerous pattern in everyday environments. Transparent glass walls — found in schools, airports, malls, and shopping centres — are built for aesthetics. They make spaces look modern, open, and clean. But their greatest strength is also their greatest weakness: they are almost invisible.

In schools, students rushing between classes often fail to notice the glass in front of them.

In airports, passengers distracted by screens or luggage walk into clear partitions.

In malls and shopping centres, reflective floors, bright interiors, and heavy foot traffic make the glass blend perfectly with the surroundings.

The problem was not behaviour.

The problem was visibility.

We broke the issue down into two clear causes:

1. Transparent Glass Becomes Invisible During Motion.

When a person is walking or running — especially indoors — their focus shifts forward, not sideways. With no colour, pattern, or contrast, the glass disappears into the background.

2. Modern Distractions Reduce Awareness Even Further.

In public spaces, people use their phones, talk in groups, or hurry to the next task. Even adults in airports or malls experience sudden collisions because they simply didn't see the glass in time.

With this understanding, VoltEdge built the system around a single principle:

make the glass visible only when it needs to be visible.

This led to our three-layer system design:

1. Input Layer — Detecting Approach

We integrated:

- Ultrasonic Sensor (HC-SR04) to detect the distance of the approaching individual
- PIR-based motion logic to interpret walking vs. running

These sensors give the system awareness — the ability to recognise when someone is getting close to the glass and how fast.

2. Processing Layer — Analysing Behaviour

At the centre of the system is the Arduino UNO, which continuously processes:

- distance readings
- motion triggers
- speed changes

This allows the system to classify:

- safe walking
- cautious approach
- or fast, risky movement

The microcontroller decides exactly when and how strongly the alert should activate.

3. Output Layer — Warning the User

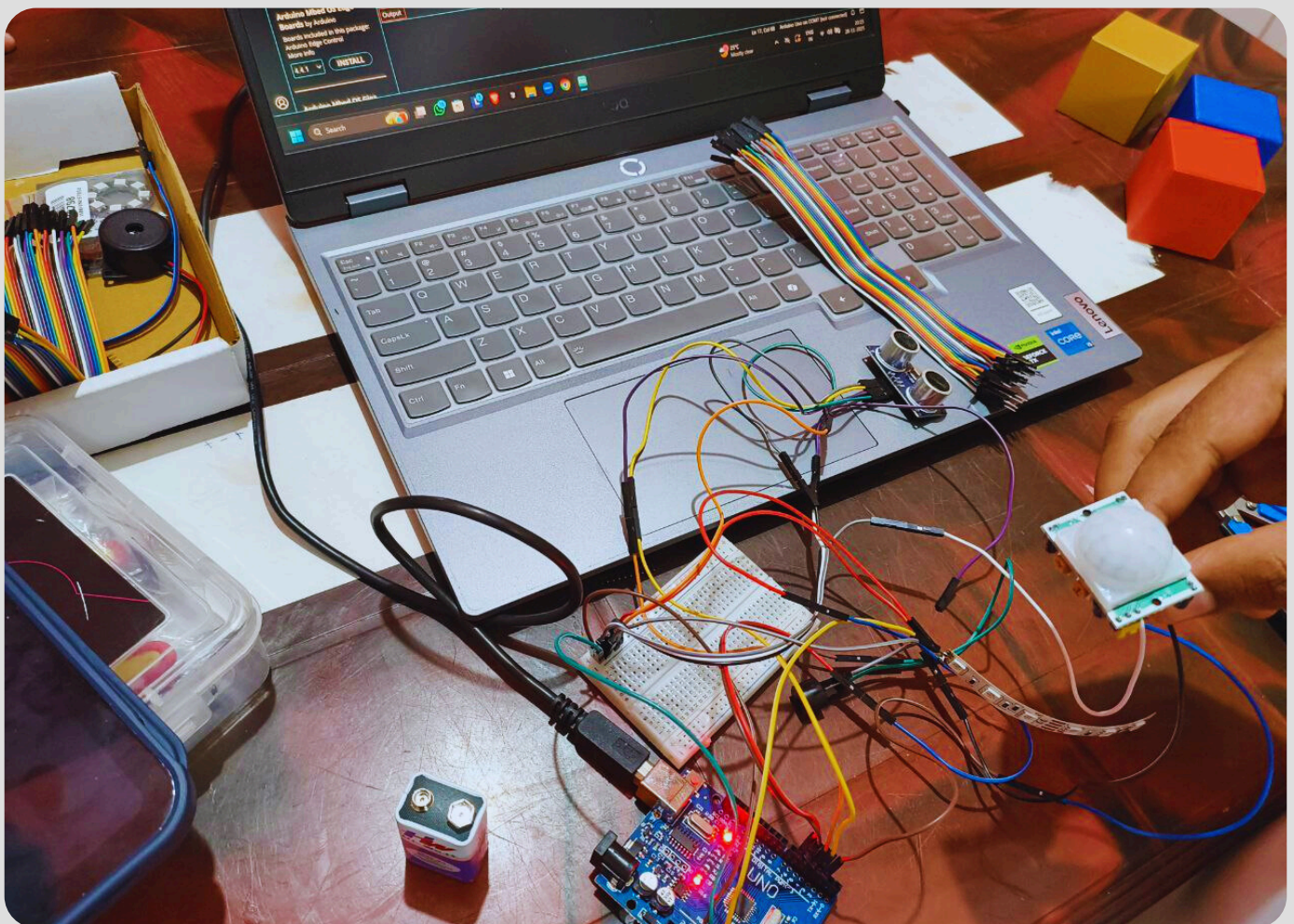
To make the glass visible only when needed, we used:

- LED Strip illumination to highlight the edges of the glass
- Active Buzzer for an audible warning

The response changes based on speed:

- A soft LED glow for slow walking
- A brighter, instant light + louder buzzer for fast movement

This dual alert ensures that even distracted individuals — especially in public spaces — notice the transparent wall and avoid collision.



Implementation & Technical Architecture

Transforming the Smart Glass Safety Wall from an idea into a functioning system required the same disciplined engineering mindset we apply in robotics. VoltEdge approached the project with a layered, structured architecture that allowed each part of the system to build naturally upon the next. This method ensured reliability and accuracy in real-world environments such as schools, airports, malls, and shopping centres — places where transparent glass often becomes invisible, leading to frequent accidents.

The foundation of the system lies in a compact hardware setup built around components we selected deliberately for both reliability and accessibility. At its core, an Arduino UNO acts as the controller, receiving signals and executing decisions. An ultrasonic sensor supplies distance measurements, giving the system a sense of how close a person is to the glass surface. A PIR motion sensor complements this by detecting motion patterns, helping the system determine whether someone is calmly walking or approaching at higher speed. Along the edges of the glass, we integrated a high-visibility LED strip that activates according to the threat level, paired with an active buzzer that provides auditory alerts. These components sit alongside supporting wiring, power regulation, and a diffuser film to ensure the lighting spreads smoothly across the glass frame — all arranged in a discreet configuration that preserves the visual integrity of the space.

The behaviour of the entire setup is governed by Arduino logic structured into a sequence of sensing, decision-making, and action. The first module continuously reads real-time data from the ultrasonic and PIR sensors, giving the system constant awareness of its surroundings. The next layer evaluates this input and interprets it into meaningful categories: normal walking, caution-level proximity, or a high-risk fast approach. Based on this classification, the system activates its outputs with precise intensity — from a gentle LED glow for safe approaches to bright illumination and a strong buzzer response for high-speed movement. This adaptive behaviour ensures that the system reacts not just to presence, but to intent and speed.

VoltEdge designed the architecture in three simple but effective layers. The input layer, made of the ultrasonic and PIR sensors, serves as the “eyes and ears” of the system. The processing layer — the Arduino — functions as the brain, constantly evaluating conditions. The output layer, comprised of LEDs and a buzzer, communicates instantly and unmistakably with the approaching user. Despite the simplicity, the three-layer structure offers clarity, scalability, and ease of installation in public environments.

Once assembled, the system underwent iterative testing on a mock glass frame to refine its accuracy. We validated distance readings across different ranges, observed motion detection for both walking and running, tested light visibility under varied lighting conditions, tuned the buzzer’s intensity to match urgency levels, and evaluated the combined response during rapid approach scenarios. Each round of testing allowed us to fine-tune thresholds until the system behaved reliably and consistently across all conditions.

In its final form, the Smart Glass Safety Wall follows a clear behaviour cycle: detect movement, analyse approach speed, and respond with the appropriate visual and auditory alert. Through simple components paired with intelligent logic, VoltEdge created a solution that enhances safety around transparent glass while preserving the architectural beauty of public spaces — a practical example of engineering serving everyday life.

Team Journey Summary

Complete Journey Timeline

4 October — The First Spark

Five days before midterm exams, Omer reached out to Abdullah and shared the details of the National Robotics League. The idea of competing at IIT Bombay lit a spark instantly. Even with exams around the corner, the team didn't hesitate — the possibility felt bigger than the pressure.

9 October — Official Registration

One day before midterms, the team made the decision that defined everything that came after: VoltEdge registered for NRL 2025. There was no robot yet, no parts, no strategy — just belief.

20 October — The BaseBot Arrives in Bhatkal

The kit finally reached town. Opening the box felt like the real beginning. The parts were new, unfamiliar, and scattered — but for VoltEdge, it felt like opportunity.

22 October — The Build Begins

Right after exams, the full team began assembling the BaseBot. From chassis fitting to wiring to aligning motors, every evening turned into a build session.

This was also when misunderstandings happened but resolved quickly — proving the team was stronger together.

22–27 October — Continuous Build Week

For five straight days, the team worked after classes and through the evenings.

- Ahmed Irfan led mechanical assembly
- Mohammed Zaid worked on electronics
- Abdullah attempted BaseBot programming for the first time
- Omer refined controller logic later

By 27 October, the BaseBot was completely assembled — a full prototype ready for testing.

6 November — The Robotics Workshop at Ali Public School

VoltEdge conducted its first outreach event. Nearly 60 students attended. From sensors to timelines to live demonstrations, the workshop became one of our most important contributions to STEM in our town.

6 November — First NRL Weekly Challenge Announced (Team Reveal)

The challenge arrived the same day as the workshop. With the team preparing for a tour, we shot the team reveal early the next morning.

7 November Morning — Team Reveal Shoot

Before leaving for their educational tour, the team shot the team reveal video in the school campus.

At 11:15 PM, the members left for the tour.

7–13 November — Editing Done in Isolation

While the team was away, Abdullah carried the entire responsibility of editing the team reveal video alone.

This became one of the defining contributions to VoltEdge's online identity.

13 November — Team Returns, Work Resumes

As soon as the team returned, testing, wiring corrections, and programming refinements continued.

15–25 November — Continuous Testing & Autonomous Work

Evening sessions became the norm.

The team tested motor behaviour, servo angles, gripper response, and began integrating their self-purchased IR, gyro, and ultrasonic sensors. During this time, Danish Gawai Sir guided the team through the most complex autonomous logic challenges.

21–27 November — Social Media Growth Phase

VoltEdge began posting behind-the-scenes clips, apparel reveals, workshop highlights, and memes.

By 28 November, the account crossed 64,000 views in a month — the strongest digital push in our journey.

29–30 November — Robot Reveal Trailer Scheduled

VoltEdge prepared the first trailer for EdgeBot, timed strategically before portfolio submission.

4-5 December — Full Robot Reveal Video

The complete EdgeBot reveal is planned to go live right before the team prepares for their travel and the final event.

30 November — Engineering Portfolio Submission

The deadline that holds together every section, every test, every screenshot, every design decision.

VoltEdge submits its first official engineering portfolio.

6–7 December — NRL IIT Bombay (Event Week)

The final destination.

The arena where everything we learned — every late evening, every fix, every misunderstanding resolved, every test, every decision — will come together.

This timeline is not just a sequence of dates.

It is the story of how VoltEdge was built — moment by moment, test by test, decision by decision.

Obstacles, And Busy Schedule Challenges

The journey of VoltEdge was shaped as much by pressure as by progress. Every advancement we made had to fit into a month already overflowing with academic deadlines, council responsibilities, robotics commitments, and daily student life. The true challenge was not building the robot—it was building it through everything else happening at the same time.

The first challenge appeared early on 4 October, when we discovered NRL just five days before midterm exams. With unfinished chapters to revise and multiple assignment deadlines approaching, we still registered on 9 October, one day before exams began. It was a brave decision, but it meant robotics would now run parallel to academic stress. The BaseBot kit arrived on 20 October, and by 22 October, the full build began even as we continued submitting pending assignments. Every evening from 22 to 27 October was spent assembling the robot piece by piece until the BaseBot was fully completed.

On 6 November, VoltEdge conducted a robotics workshop at Ali Public School, attended by nearly 40 students. Preparing the content, planning the flow, setting up equipment, capturing media, and conducting sessions took significant effort—especially because it happened on the same day NRL released the first weekly challenge: the Team Reveal. The next morning, 7 November, the team filmed the reveal video before several members left for an educational tour at 11:15 PM, remaining away until 13 November. During this entire period, Abdullah handled all editing and coordination for the challenge alone.

Once the team returned, the workload didn't reduce. From mid-November onward, Mohiddin, Zaid, and Omer were responsible for conducting daily afternoon robotics classes, teaching sensor basics, circuits, and robot foundations to younger students. Balancing teaching responsibilities with robotics testing and school work added another layer of pressure. Meanwhile, autonomous development began, requiring us to purchase our own IR sensors, gyro, and ultrasonic modules. We spent multiple evenings testing line tracking on black tape, stabilising turns, arranging sensor placement, and debugging logic—with continuous guidance from Danish Gawai Sir when challenges became too complex.

On 25 November, the Our College Management approved our fest proposal, with the event scheduled for 2 December, placing robotics directly beside council meetings, planning work, budgeting, and department assignments. At the same time, we still needed to manage publishing reels, posters, workshop clips, and behind-the-scenes content to maintain our digital engagement, which crossed 64,000 views by 28 November. Editing and scheduling posts during chaotic days required silent, consistent effort.

On 28 November, VoltEdge completed and tested the Innovation Challenge prototype—the Smart Glass Safety Wall system. This test had to be done between autonomous trials, portfolio writing, and fest planning, adding yet another deadline to an already overloaded week.

Through all of this—assignments, tours, daily robotics classes, workshop management, autonomous testing, innovation development, social media responsibilities, and council duties—the team never stopped working. Every member filled in when someone else was unavailable, every misunderstanding was resolved quickly, and every challenge became another reason to push forward.

These challenges did not interrupt our journey. They defined it, shaped it, and proved that VoltEdge was built not just with skills—but with discipline, cooperation, and resilience.



Team VoltEdge

Representing at NRL 2025

Team ID: 007

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